



DEPARTMENT OF LABOR AND EMPLOYMENT

Division of Oil and Public Safety

LIQUEFIED PETROLEUM GAS (LPG) REGULATIONS

7 CCR 1101-15

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

ARTICLE 1 GENERAL PROVISIONS

Section 1-1 Basis and Purpose

The basis and purpose of these regulations is to: adopt nationally-recognized codes and standards; to add or clarify terminology; to add or clarify the duties of LPG facility owners, delivery drivers and certain users; and to improve the effectiveness of the division's LPG program.

Section 1-2 Technical Rationale

The technical requirements of these regulations are generally accepted as national and international codes and standards governing the minimum levels of acceptability for the design, construction, location installation and operation of equipment for storing, handling, transporting, dispensing and utilizing LPG. The adoption of these consistent standards is necessary for the preservation of the public health, safety, and welfare of the citizens of Colorado.

Section 1-3 Statutory Authority

The amendments to these regulations are created pursuant to Sections 8-20-302, 8-20-402 and 8-20-405 of the Colorado Revised Statutes.

Section 1-4 Effective Date

These amended regulations shall be effective on November 30, 2024. The previous versions of these regulations were effective September 30, 2017, May 1, 2017, February 1, 2011, and September 1, 2005.

Section 1-5 Codes Incorporated by Reference Section

1-5-1 Codes Incorporated by Reference The following codes are incorporated by reference:

- (a) NFPA 58, Liquefied Petroleum Gas Code, 2024 edition.
- (b) NFPA 54, National Fuel Gas Code, 2024 edition.
- (c) NFPA 30A, Code for Motor Fuel Dispensing Facilities & Repair Garages, 2024 edition.
- (d) NFPA 59, Utility LP-Gas Plant Code, 2024 edition.
- (e) NFPA 1192, Standard on Recreational Vehicles, 2021 edition.
- (f) NIST Handbook 44, Specifications, Tolerances, and Other Technical Requirements for Weighing and Measuring Devices, 2024 edition.

- (g) NIST Handbook 130, Uniform Laws and regulations in the Areas of Legal Metrology and Engine Fuel Quality 2024 edition.

All NFPA codes incorporated by reference may be purchased from:

National Fire Protection Association
1 Batterymarch Park
Quincy, Massachusetts
USA 02169-7471
1-800-344-3555
<http://www.nfpa.org>

All NIST codes incorporated by reference may be purchased from:

National Institute of Standards and Technology
100 Bureau Drive
Gaithersburg, MD. 20899
301-975-4004
<http://www.nist.gov>

Or for free download:

<https://www.nist.gov/pml/owm/publications/nist-handbooks>

Section 1-5-2 Inspection of Incorporated Codes

Interested parties may inspect and obtain copies of excerpts of the referenced incorporated materials for a reasonable charge by contacting the Division of Oil and Public Safety at

633 17th Street, Suite 500
Denver, CO 80202
303-318-8525
cdle_ops@state.co.us

Section 1-5-3 Later Amendments not Included

These regulations do not include later amendments to or editions of the incorporated material.

Section 1-6 Definitions

Terms in these regulations shall have the same meaning as those found in Title 8, Article 20 and Title 9, Article 4 of the Colorado Revised Statutes. In addition, unless the context otherwise requires:

AHJ -- Authority Having Jurisdiction

API – American Petroleum Institute.

ASME International – Formerly the American Society of Mechanical Engineers (ASME).

ASTM International – Formerly the American Society for Testing and Materials.

CETP — Certified Employee Training Program

Condemned – A **condemned container assembly and piping system** is one determined by a state inspector to be so unsafe that further use is prohibited until it is satisfactorily repaired or replaced.

Container assembly – A **container assembly** includes US Department of Transportation (DOT) and ASME containers, commonly known as tanks or cylinders.

CRS – Colorado Revised Statutes.

Cylinder -- A portable container (usually DOT) with a marked water capacity of 1000 lb (454 kg) or less, designed to transport and store LP-Gas.

Division – The **Division of Oil and Public Safety**, the regulatory agency of the Colorado Department of Labor and Employment having jurisdiction over LPG container assemblies and piping systems as defined in these regulations.

DOT – United States Department of Transportation.

Existing installation – Any LPG container assembly and piping system that has been placed into service and received its initial inspection by a state inspector.

FSA – Fire Safety Analysis, also known as *Product Release Prevention and Incident Preparedness Review or Incident Prevention Review*. An FSA is a plan incorporating the various safety features used to control the product and operations at the facility, an evaluation of hazard to the immediate neighborhood and a tool to be used by an emergency response agency, such as the local fire department.

Handling – Transferring LPG into a DOT container or an ASME tank. Handling does not include transporting of LPG.

Incident – A reportable accident, as defined by 8-20-407(1), CRS.

Incident prevention review – See definition under *FSA*.

Installer – Person or company responsible for setting up for use any container assembly and piping system required by LPG statute or regulation to be inspected.

Interruption of service – An interruption of gas service where pressure in the fuel gas piping is below a point that the appliances can no longer operate, including an out-of-gas call, except for when conducting a test.

LPG - Liquefied Petroleum Gas (LP-Gas). Any material having a vapor pressure not exceeding that allowed for commercial propane that is composed predominantly of the following hydrocarbons, either by themselves (except propylene) or as mixtures: propane, propylene, butane (normal butane or isobutane), and butylenes.

LPG facility – Facility that has an LPG container assembly and piping system.

LPG liquid meter – A meter designed to measure LPG liquid, with an active National Type Evaluation Program (NTEP) Certificate of Conformance (CC), which includes a metering chamber or device, vapor eliminator, differential valve and register.

Marking – Container information located on container name-plate or stamped into the shell or collar of an LPG container. Marking is done in accordance with the regulations of DOT, ASME or API-ASME and is done by the original container manufacturer or a repair organization authorized by the National Board to utilize the “R” code symbol stamp.

NACE – The National Association of Corrosion Engineers.

National Board – The National Board of Boiler and Pressure Vessel Inspectors.

New installation – Any container assembly that has been placed into service but has not received its initial inspection by a state inspector.

NFPA – The National Fire Protection Association.

NIST – The National Institute of Standards and Technology. Out-of-gas call – A request for LPG delivery to an empty tank. PEP — PERC Education Program

PERC – The Propane Education and **Research** Council.

Plant Oil Extraction – Process using LPG as a solvent for the isolation of oil from raw plant matter.

Plant Oil Extraction Equipment – Any equipment used for the purpose of plant oil extraction as defined.

Product release prevention and incident preparedness review – see definition under **FSA**.

Proved – The act of having verified the accuracy of meters used to measure fuel and petroleum products using a prover.

Prover – A calibrated volumetric receiver or mechanical device traceable to NIST standards.

PSI – Pounds per square inch.

Registered serviceperson – Any individual who, for hire, award, commission or any other payment of any kind, installs, services, repairs or reconditions a commercial weighing or measuring device and who voluntarily registers with the division.

Registered service agency (RSA) – Any agency, firm, company or corporation that, for hire, award, commission or any other payment of any kind, installs, services, repairs or reconditions a commercial weighing or measuring device and voluntarily registers with the division. Under agency registration, identification of individual servicepersons shall be required.

Retail – The sale of LPG from fixed dispensing equipment or by means of bobtail delivery truck, or such as at a fueling station, in small or individual quantities for direct consumption by the purchaser.

State inspector – A person who is employed or authorized by the division to perform inspections of LPG facilities.

Temporary Installation – A once only installation meeting all requirements of NFPA 58, in service no longer than 12 months.

Universal Cylinder - A cylinder (usually DOT) that can be connected for service in either the vertical or the horizontal position so that the fixed maximum liquid level gauge, pressure relief device, and filling and withdrawal appurtenances function properly in either position.

Section 1-7 Applicability

The regulations contained herein shall apply to the operation of all LPG container assembly and piping systems including the following:

- (a) Containers, piping and associated equipment, when delivering LPG to a building for use as a fuel gas.
- (b) Pipeline terminals, natural gasoline plants, refineries, tank farms, underground storage facilities, aboveground storage facilities and chemical plants utilizing LPG in the manufacture of their products.
- (c) The design, construction, installation and operation of pipeline terminals that receive LPG from pipelines under DOT jurisdiction, whose primary purpose is the receipt of LPG for delivery to transporters, distributors or users. Coverage shall begin downstream of the last pipeline valve or tank manifold inlet.

Section 1-8 Condemning an LPG Container Assembly and Piping System

- (a) Conditions which a state inspector may determine to be unsafe include: bypassed safety controls; inoperative relief valves; any gas leak from an LPG container assembly; any excessive gas leak from the piping system; missing nameplate; or any other condition deemed by a state inspector to be unsafe. A container assembly or piping system that meets any condition described above may be condemned by a state inspector.
- (b) The owner or user must shut down the condemned LPG container assembly and piping system as directed by a state inspector. If neither the owner nor user is available, a state inspector will cause the system to be shut down.
- (c) A state inspector will affix a notice to a condemned LPG container assembly and piping system stating that it has been condemned and may not be used until satisfactory repairs are made, as determined by a re-inspection by a state inspector or other person authorized by the division.

ARTICLE 2 INSTALLATION

Section 2-1 General Requirements

- (a) All new LPG installations shall be constructed in accordance with the incorporated editions of the NFPA and NIST codes.
- (b) All existing LPG installations shall be constructed in accordance with the incorporated editions of the NFPA and NIST codes in effect at the time of construction, including any retroactive requirements adopted by the division.

Section 2-2 Installation Permits

- (a) Plans for all installations utilizing LPG storage containers of over 2,000 gallons water capacity shall be submitted to the division for approval before construction, including installation, replacement or relocation of such installations, begins.

- (b) The division shall be notified of temporary installations of the container sizes covered in (a) before the installation is started.
 - (1) Notification shall be accomplished using the process and the permit application form listed in Section 2-2 Installation Permits (d)
 - (2) Pre-approval shall be obtained from the Division for any temporary installation that is in use for more than twelve months.
- (c) Plans for any of the following shall be submitted to the division for approval before installation:
 - (1) Service stations supplying LPG for motor fuel.
 - (2) Installations for filling of DOT and ASME mobile containers and containers marked to demonstrate compliance with Federal Aviation Administration regulations.
 - (3) Industrial bulk storage installations and all other bulk storage installations utilizing storage containers for LPG of over 2,000 gallons aggregate water capacity.
 - (4) LPG liquid transfer operations using cylinders to fill other cylinders for use in plant oil extraction per 6-1 (b) of these regulations.
- (d) The permit procedures are as follows:
 - (1) The permit application shall be submitted on an application form approved and provided by the division. The application form may be found at:
<https://ops.colorado.gov/Petroleum/LiquefiedCompressedGases/LiquefiedPetroleumGases>.
 - (2) The application shall include a plot plan containing all elements required by the division.
 - (3) The division may deny the application if the proposed installation does not conform to the division LPG statute or regulation or to codes adopted by the division, or if the application is incomplete or determined to be inaccurate.
 - (4) Construction and installation of tank and piping shall conform to code(s) in effect at the time of installation.
 - (5) The division may revoke a permit if construction is not performed per the approved permit, or if the construction fails to meet operating or fire safety regulations established by the division or by the applicable NFPA Code.
 - (6) An installation permit approved by the division is automatically revoked if construction does not begin within 6 months of approval, unless a written request for an extension is submitted to and approved by the division.
 - (7) For new installations with an aggregate over 4,000 gallons, an FSA must be in effect prior to the operation of the installation per NFPA 58, following guidance from Annex A, A.6.29.2 and A.6.29.3 or another nationally-accepted standard approved in advance by the division.
 - (8) A local authority having jurisdiction, including fire departments, may require and enforce more stringent requirements than these regulations.

Section 2-3 Access Requirements

- (a) The division may inspect an LPG facility at any time during its construction. Access shall be provided to the division or its agent for such purpose upon request.
- (b) After an LPG container has been installed, the division may inspect the container to verify compliance with design, construction, location, installation and operation requirements.
- (c) LPG installers, facility owners, tank owners and owners of locations where an LPG container is installed shall notify the division once the installation is completed and grant inspection access to the division or its agent for such purpose upon request.

Section 2-4 Corrosion Prevention Requirements

- (a) Corrosion protection is required for all underground steel LPG tanks and piping installed after September 1, 2005, and for all other underground steel LPG tanks by January 1, 2011.
- (b) Corrosion protection shall meet the requirements of NFPA 58.
- (c) Cathodic protection systems shall be installed and tested in accordance with the provisions of NFPA 58, and documentation of the results of the two most recent tests shall be retained.
- (d) Installers of cathodic protection systems shall be trained in the proper installation of cathodic protection systems, using the PERC Cathodic Protection Guide, or PEP equivalent. Training shall be documented per Section 4-1 of these regulations, and refresher training shall be required at least every three years.

Section 2-5 Nameplates

- (a) When an ASME container assembly has a nameplate with a legible serial number but has insufficient data to be in service, and there is a Manufacturer's Data Report (U1A) with a matching serial number and the container fits the description on the U1A, the owner/operator may then make application to the division for a tag bearing the manufacturer's name, ASME Code symbol, maximum allowable working pressure, UG or AG (or both AG and UG) and water gallon capacity. If granted, the division shall issue an identifying tag to be permanently attached to the container by a state inspector.
- (b) When an ASME container assembly has no nameplate but there is sufficient data to prove that it once had a nameplate showing the critical data thereon, the owner/operator may then make application to the division for a tag bearing the manufacturer's name or trademark, ASME Code symbol, maximum allowable working pressure, UG or AG (or both AG and UG) and water gallon capacity. If granted, the division shall issue an identifying tag to be permanently attached to the container by a state inspector. A copy of the container's U1A form shall be considered important data.
- (c) When a container assembly and piping system has a nameplate whose stamping becomes indistinct, or the nameplate is lost or illegible but traceability to the original container assembly item is still possible, the owner or user shall have the stamped data replaced as follows:
 - (1) All re-stamping shall be done in accordance with the version of the code in effect at the time of container construction.
 - (2) A request for permission to re-stamp or replace a nameplate shall be made in advance to the division. Proof of the original stamping and the manufacturer's data report shall be furnished with the request.

- (3) Permission from the division is not required for the reattachment of nameplates that are partially attached.
 - (4) Re-stamping or replacement authorized by the division shall be witnessed by a state inspector.
 - (5) The stamping shall be identical to the original stamping. When the Code symbol is to be re-stamped, it shall be done by the original manufacturer and witnessed by a state inspector.
 - (6) Replacement nameplates shall be clearly marked "replacement."
 - (7) After replacing a nameplate, the owner or user shall file with the division a copy of the stamping or nameplate as applied and shall include the signature of the state inspector who witnessed the replacement.
- (d) If replacement of the nameplate is not possible because the container assembly cannot be traced, a hydrostatic test or other test approved by the division must be performed if the container is to remain in service.
- a. Such test shall be performed by an independent contractor experienced in hydrostatic tests. The test shall be conducted according to the procedure outlined in ASTM International Designation: E 1003 – 95 (Reapproved 2000) at 1.5 times working pressure ($250 \text{ psi} \times 1.5 = 375 \text{ psi}$) to evaluate the integrity of the container.
 - b. The owner/operator may then make application, including the results of the hydrostatic test, to the division for an exception. If granted, the division shall assign a number to the container and issue an identifying tag with that number to be permanently attached to the container by a state inspector.
 - c. With the division's approval, the container may continue to operate at that location indefinitely. The container shall not be moved and reinstalled at any location, including elsewhere at the same facility.
- (e) If a container was installed prior to September 1, 2005, and is missing the nameplate, one of the following shall be performed:
- a. The nameplate shall be replaced per Section 2-5-1 (a) of these regulations. A hydrostatic test or other test approved by the division shall be performed and approval to operate the container granted by the division per Section 2-5 (b) of these regulations.
 - b. The container shall be permanently removed from service.

Section 2-6 Fire Safety Analysis (FSA)

- (a) For all new LPG installations in excess of 4,000 gallons in aggregate an FSA is required to be completed by the operational date of the installation.
- (b) The FSA shall be prepared in accordance with the requirements of NFPA-58, including Annex A, A.6.29.2 and A.6.29.3 or any other nationally-accepted standard approved by the division in advance.

- (c) It is not required that the FSA be prepared or approved by a professional engineer: however, the preparation should be completed by someone who, at a minimum, is familiar with the properties of LPG, the application of NFPA 58 and the physical layout of the installation. The preparer shall consult with the local fire protection district to complete the FSA. Some modifications to the installation may require the services of a registered professional engineer.
- (d) The most current FSA document shall be maintained at the LPG installation, where it shall be available for inspection by the division upon request.

Section 2-7 Marking Underground Containers in Snow Areas

- (a) In areas where snow can be expected to cover the dome lids of underground containers, such containers shall be marked so that emergency and LPG service company personnel can locate the tank for emergency shut down purposes or to service the tank. Such marking shall be accomplished by placing a stake or other marking that shall be installed higher than the anticipated maximum snow level up to a height of 15 feet.

ARTICLE 3 DELIVERY AND DISPENSING

Section 3-1 LPG Delivery

- (a) Containers shall be filled only after determination from the point of transfer that the installation of the container and visible exterior piping system comply with the provisions of NFPA 58 and these regulations.
- (b) It is permissible to fill an ASME tank whose nameplate or other marking is damaged, provided the following information can be determined:
 - (1) If the container is registered with the National Board and the facility can produce the Manufacturer's Data Report (U-1A and/or U-2A) form, the manufacturer's name and container's serial number must be legible.
 - (2) If the container is not a National Board-registered container, the manufacturer's name, the container's serial number, the ASME stamp and the pressure rating must be legible.
- (c) If the nameplate or other marking is missing, delivery to the tank is prohibited. Refer to Section 2-5 of these regulations for instructions on re-attaching or re-placing nameplates.
- (d) Delivery to a container with corrosion that appears to be greater than 1/3 of the thickness of the metal is prohibited.
- (e) When noncompliance with the provisions of the incorporated codes is found, the container owner or user of record shall be notified in writing. Section 3-2 Delivery after Interruption of Service
 - (a) When delivery is made to an LPG container assembly that has had an interruption of service as defined by these regulations, a leak check shall be performed immediately after LPG is introduced into the piping.
 - (b) The leak check shall be performed according to one of the methods set forth in Annex D (Suggested Method of Checking for Leakage) of NFPA 54 or another method approved by the division in advance
 - (c) The person performing the leak check shall document that the test was performed. The documentation shall include, at minimum, the following information:

- (1) Date the test was performed
 - (2) Test start time
 - (3) Test end time
 - (4) Name of person performing the test
 - (5) Name of person's employer
 - (6) Address and phone number of person's employer
 - (7) Type of test
 - (8) Test start pressure and end pressure if a constant pressure is used
- (d) Documentation of the leak check shall be retained for a minimum of one year by the employer of the person who performed the test.

Section 3-3 Dispensing

- (a) The requirements of Section 3-4 of these regulations shall apply to dispensers at retail facilities.
- (b) All retail and non-retail LPG dispensers must comply with the minimum standards as prescribed by the applicable sections of the incorporated codes.
- (c) All LPG sales shall be offered, exposed for sale, or sold by one of the following.
- (1) Weight- by the kilogram or pound.
 - (2) Liquid Volume- by the liter (defined as 1 liter @ 15°C) or the gallon (defined as 231 in³ at 60° F).
 - (3) Flat Rate- (defined as the act of filling a customer's cylinder to the maximum allowable capacity).

Section 3-4 Retail Motor Fuel and Fuel Gas Dispenser Meter Inspection and Testing

- (a) All retail LPG dispensers shall be suitable for their intended use, properly installed and accurate, and they shall be maintained in that condition by their owner/operator.
- (b) All retail LPG dispensers shall have an active National Type Evaluation Program (NTEP) Certificate of Conformance (CC) prior to their installation or use for commercial purposes.
- (c) The division shall be notified when any new or remanufactured retail LPG dispenser is placed in service at a new or existing installation. Notification shall be submitted using a placed in service report provided by the division.
- (d) All metered sales by liquid volume shall be accompanied using a metering system as follows:
- (1) Sales using metering systems with a maximum rated capacity greater than 20 gal/min shall be accomplished using a metering system that automatically compensates for the effects of temperature.

- (2) Sales using metering systems with a maximum rated capacity equal to or less than 20 gal/min that were placed into service after January 1, 2026, shall be accomplished by use of a metering system that automatically compensates for the effects of temperature.
 - (3) Effective January 1, 2030, all metered sales (through all capacities of metering devices, regardless of installation and service date) shall be accomplished by use of a metering system that automatically compensates for temperature.
- (e) No owner/operator of any retail LPG dispenser shall use the dispenser for the measurement of LPG unless it has been proved in a manner acceptable to the Director of the division and sealed as correct by a state inspector or registered service agency.
- (f) If any retail LPG dispenser fails to comply with any of the provisions of this regulation, a state inspector shall seal it in such a manner as to prohibit its use, and it shall remain sealed until it complies with all of the provisions of this regulation.
 - (1) When a retail LPG dispenser is brought back into compliance with these regulations, it must be placed back in service by a state inspector or registered service agency.
- (g) All retail LPG dispensers shall comply with the minimum standards as prescribed by the applicable sections of the incorporated codes, except as modified or rejected by these regulations or by the director.
- (h) All retail LPG dispensers shall be labeled in accordance with the minimum standards as prescribed by the applicable sections of the incorporated codes except as modified or rejected by these regulations or by the division.

Section 3-5 Bobtail Delivery Truck Meter Inspection and Testing

- (a) All LPG bobtail delivery truck metering systems shall be suitable for their intended use, properly installed and accurate, and they shall be maintained in that condition by their owner/operator.
- (b) All LPG bobtail delivery truck metering systems shall have an active National Type Evaluation Program (NTEP) Certificate of Conformance (CC) prior to their installation or use for commercial purposes.
- (c) The division shall be notified when any new or remanufactured LPG bobtail delivery truck metering systems is placed in service at a new or existing installation. Notification shall be submitted using a placed in service report provided by the division.
- (d) No owner/operator of any LPG bobtail delivery truck metering system shall use the meter for the measurement of LPG unless it has been proved in a manner acceptable to the Director of the division and sealed as correct by a state inspector or registered service agency.
- (e) If any LPG bobtail delivery truck metering systems fails to comply with any of the provisions of these regulations, a state inspector shall seal it in such a manner as to prohibit its use, and it shall remain sealed until it complies with all of the provisions of these regulations.
 - (1) When an LPG bobtail delivery truck metering system is brought back into compliance with these regulations, it must be placed back in service by a state inspector or registered service agency.
- (f) All LPG bobtail delivery truck metering systems shall comply with the minimum standards as prescribed by the applicable sections of the incorporated codes except as modified or rejected by these regulations or by the division.

- (g) All retail LPG bobtail delivery truck meters shall be labeled in accordance with the minimum standards as prescribed by the applicable sections of the incorporated codes except as modified or rejected by these regulations or by the division.

Section 3-6 Filling Containers by Weight

- (a) All cylinders less than 200 pounds water capacity (i.e., 100-pound cylinders), with the exception of forklift cylinders and hot air balloon containers, shall be filled by weight.
- (b) Volumetric filling of forklift cylinders from bobtail delivery trucks shall be allowed in accordance with NFPA 58, Section 6.7 and Table 6.7.2.1, and all personnel shall be trained in proper handling procedures in accordance with NFPA 58, Chapter 7.
- (c) Scales used for filling LPG containers must be inspected annually and found to be in compliance with the specifications and tolerances published in NIST Handbook 44. This certification must be performed either by the Colorado Department of Agriculture or by a person authorized by the Colorado Department of Agriculture. Any necessary repairs must be completed within 30 days.
- (d) If the Colorado Department of Agriculture fails to perform the annual inspection in a timely fashion, the scale may remain in operation, provided the scale owner has not prohibited or hindered such inspection by the Colorado Department of Agriculture, and further provided that any repairs required at the most recent previous inspection have been completed.

Section 3-7 Tank Delivery Truck, Semitrailer, or Truck Trailer for Bulk Storage.

- (a) No tank delivery truck, semitrailer, or truck trailer shall be used as a bulk storage plant for LPG unless the same has been permitted, inspected and approved as a temporary installation by the division.
- (b) Where cargo tank vehicles are filled from other cargo tank vehicles or cargo tanks, the following requirements shall apply:
 - (1) Transfer between cargo tanks or cargo tank vehicles where one is used as a bulk plant shall be temporary installations that comply with NFPA 58, 4.3.2, 6.18.1, 6.18.2, 6.18.4 through 6.18.6, and 7.2.3.1. through 7.2.3.2. and 7.2.3.9.
 - (2) Arrangement and operations of the transfer system shall be in accordance with NFPA 58 Table 6.5.3.

ARTICLE 4 TRAINING REQUIREMENTS

Section 4-1 General Training Requirements

- (a) Any person who transfers LPG including, but not limited to, dispenser operators, bobtail delivery drivers and transport operators, or who services or installs exterior piping of LPG vapor distribution systems, shall receive adequate training to perform all related duties safely and in accordance with the provisions of NFPA 58, Sections 4.4 and 7.2.
- (b) The employer of any person referenced in (a) above, shall document that person's training and shall make these records available to the division or its agent upon request. The records shall include the following information, at minimum:
 - (1) Person's name
 - (2) Training date(s)

- (3) Name of trainer
- (4) Topics covered by training
- (5) Verification by the person's supervisor or certification described in this Article that the person has demonstrated adequate knowledge and skill to perform assigned duties
- (c) For all training required under Sections 4-2 through 4-4 of these regulations, refresher training shall be required at least every three years. The training shall be documented.
- (d) The employer of the person who received the training shall maintain that documentation as long as the person remains an employee.

Section 4-2 Dispenser Operator Training

- (a) The minimum training requirements for dispenser operators, that shall be completed prior to operating LPG dispensers, may be satisfied by certification by one of the following:
 - (1) Certified Employee Training Program (CETP) *Basic Principles and Practices*, or PEP equivalent as published by PERC.
 - (2) A training program which contains, at minimum, certification by the *Dispensing Propane Safely*, or PEP equivalent as published by PERC.
 - (3) Equivalent training program approved by the AHJ.

Section 4-3 Delivery Personnel Training

- (a) The minimum training requirements for delivery personnel shall include:
 - (1) Proper procedure for filling an ASME container
 - (2) Knowledge of when a leak check is required
 - (3) Proper procedure for conducting and documenting a leak check
 - (4) Criteria for determining when filling a container is prohibited because of improper installation or because of excessive corrosion, dents or gouges
 - (5) Emergency procedures as outlined in the employer's FSA
 - (6) Completion of the following training programs published by PERC
 - (A) CETP *Basic Principles and Practices*, or PEP equivalent.
 - (B) CETP Bobtail Delivery Operations, or PEP equivalent.
 - (C) CETP *Basic Plant Operations*, or PEP equivalent.
- (b) All required training for delivery personnel outlined in 4-3 (6) shall be completed within one year of assuming current job duties.

Section 4-4 Service and Installation Personnel Training

- (a) The minimum training requirements for those who service or install exterior piping of LPG vapor distribution systems may be satisfied by completing the following training programs published by PERC.
 - (1) CETP *Basic Principles and Practices*, or PEP equivalent.
 - (2) CETP *Designing & Installing Exterior Vapor Distribution Systems*, or PEP equivalent.
 - (3) CETP *Placing Vapor Distribution Systems and Appliances into Operation*, or PEP equivalent.
- (b) All required training for those who service or install exterior piping of LPG vapor distribution systems shall be completed within one year of assuming current job duties.

Section 4-5 Transport Operator Training

- (a) The minimum training requirements for transport operators may be satisfied by completing the following training programs published by PERC.
 - (1) CETP *Basic Principles and Practices*, or PEP equivalent.
 - (2) PERC *Transport Operator Training Program*, or PEP equivalent.
- (b) All required training for transport operators shall be completed within one year of assuming current job duties.

Section 4-6 Dispensing System Installer Training

- (a) Dispensing Systems installed after July 1, 2025, must be installed by personnel meeting minimum training requirements.
- (b) The minimum training requirements for dispensing system installers may be satisfied by completing the following training programs published by PERC.
 - (1) CETP *Basic Principles and Practices*, or PEP equivalent.
 - (2) CETP *Designing & Installing Dispenser Transfer Systems*, or PEP equivalent.

ARTICLE 5 ACCIDENT REPORTS AND INVESTIGATIONS

Section 5-1 Reportable Accidents

- (a) Reports of accidents, fires, explosions, injuries, damage to property or loss of life at installations using liquefied petroleum gas shall be reported to the division within 24 hours after their occurrence.
- (b) Subsection (a) of this Section includes accidents resulting from the improper use of equipment, appliances, and appurtenances to LPG systems. The division may investigate such occurrences and shall maintain a written record of findings, which shall be available for public examination.

Section 5-2 Reporting Requirements

- (a) The following persons are required to notify the division of an LPG accident that meets any of the criteria of Section 5-1 of these regulations:
 - (1) Owner or the owner's representative of the LPG facility, if the accident occurred at the facility
 - (2) Employer of the delivery personnel, if the accident occurred during delivery
 - (3) Employer of the delivery personnel, if the accident occurred post-delivery and the employer received notification of it
- (b) Accidents may be reported by telephone or email using the following information:
 - (1) Telephone: 303-318-8547
 - (2) Email: cdle_oil_inspection@state.co.us
- (c) The accident report shall include, at minimum, the following information.
 - (1) The names of the operator and person making the report and their telephone numbers
 - (2) The date, time, and location of the accident
 - (3) The number of fatalities and personal injuries
 - (4) All other significant facts known by the person making the report that are relevant to the cause of the accident or extent of the damages

ARTICLE 6 INSTALLATION AND USE OF CONTAINERS AND SYSTEMS FOR PLANT OIL EXTRACTION

Section 6-1 Transfer Operations

- (a) LPG liquid transfer operations using cylinders shall be subject to all applicable provisions of NFPA 58 and these regulations.
- (b) LPG liquid transfer operations using cylinders shall be subject to the permitting requirements located in Section 2-2 of these regulations.
- (c) While cylinders are connected for transferring LPG liquid into another cylinder, and they are grouped with additional cylinders consisting of a total aggregate water capacity including the connected cylinders, of greater than 501 gallons, the entire group of cylinders shall meet the minimum separation distances outlined in NFPA 58 Table 6.4.1.1 along with 6.4.3 (2-4).
- (d) LPG liquid transfer operations and points of transfer shall be located in accordance with Table 6.7.2.1 of NFPA 58.
- (e) Security shall be provided to restrict unauthorized access to LPG transfer areas.
- (f) The filling location must be located 15' from combustion engine use, i.e., vehicle parking. All ignition sources shall be shut off.
- (g) Where heating blankets are used, they must be listed for use in explosive environments.

- (h) Smoking, portable lighting, portable electric tools, etc. shall not be in use within 25' of the filling operation.
- (i) Purging of cylinders may be performed at the cylinder filling location in accordance with NFPA 58.
 - (1) The cylinder must be rapidly dispersed in the atmosphere
 - (2) Where environmental conditions do not allow rapid dispersal (i.e., wind conditions, site conditions such as neighboring buildings on lot lines not allowing adequate natural ventilation, etc.), LPG must be flared using a method in compliance with NFPA 58
- (j) LPG liquid transfer operations shall be conducted outdoors unless done in a building or room that has been designed in accordance with NFPA 58 Chapter 10 requirements and provides appropriate flammable gas detection equipment.
- (k) LPG liquid transfers using lighter refill-type containers using can taps or other puncture equipment is prohibited.
- (l) Persons employed in the transfer of LPG liquid from one cylinder to another shall be trained in accordance with the provisions of Article 4, Sections 4-1 & 4-2 of these regulations.
- (m) With the exception of a "universal cylinder", cylinders used in transferring LPG liquid shall have a withdrawal valve incorporating a "wet leg" that allows LPG liquid to be withdrawn while the cylinder is in the upright position. Transfer of LPG liquid shall not be accomplished by means of tipping or inverting a cylinder.
- (n) Fixed liquid level gauges or other "vent type" gauges, shall not be used for filling cylinders with unodorized LPG. Filling of cylinders with LPG that is unodorized shall be accomplished only by means of weight, using a scale that is in compliance with Section 3-6 (c) of this regulation, and they shall be Class I Division I listed or be of the mechanical type.
- (o) Only inert gases shall be allowed to be injected into a cylinder or other container in order to transfer LPG liquid from one container to another. Injection of compressed air, oxygen, or any oxidizing gas into containers to transfer LPG liquid shall be prohibited.
- (p) Electrical equipment is required to be Class I Division I within 5' and Class I Division II within 15' of filling operations.
- (q) While cylinders are in storage awaiting use, they shall meet all of the requirements of NFPA 58 Chapter 8, *Storage of Cylinders Awaiting Use, Resale, or Exchange*.
- (r) All containers that contain unodorized LP-Gas shall be marked in accordance with NFPA 58, Chapter 5.2.8.5.

Section 6-2 Extraction Equipment

- (a) Extraction equipment using LPG are required to be listed or approved by the local AHJ.
- (b) In lieu of listing, an engineering report (signed and sealed by a licensed Colorado engineer) must be retained on site and submitted to the local AHJ where required, demonstrating how the system meets national standards, i.e., NFPA 58, as a basis of design, including an analysis/description of every component of the system

- (c) Only closed-loop type LPG extraction equipment is permitted. Open blasting extractions or equipment that releases LPG to the atmosphere during the extraction process is strictly prohibited.
 - (d) An owner's operation manual containing specific instructions regarding proper use of equipment and any safety provisions must be retained on site with the equipment and submitted to the local AHJ where required.
 - (e) All equipment utilizing unodorized LPG must employ flammable gas detection equipment as approved by the AHJ.
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Editor's Notes

History

Entire rule eff. 02/01/2011.

Entire rule eff. 05/01/2017.

Sections 1-4, 1-5 eff. 09/30/2017.

Entire rule eff. 11/30/2024.